

Delta Live Tables (DLT) in Databricks - Beginner Guide

1. What is Delta Live Tables?

Delta Live Tables (DLT) is a Databricks framework used to build reliable data pipelines. It allows engineers to define data transformations and Databricks manages pipeline execution, dependencies, monitoring, and table updates.

Note: Databricks has evolved DLT into Lakeflow Spark Declarative Pipelines. Existing DLT concepts and code continue to work, but newer naming uses Lakeflow terminology.

2. Why use DLT?

Traditional ETL pipelines require manually managing:

- Job scheduling
- Dependency order
- Data quality checks
- Error handling
- Table refresh logic

DLT provides:

- Declarative pipeline development
- Automatic dependency management
- Data quality expectations
- Monitoring and lineage

3. DLT Architecture Flow

Source Data

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Bronze Layer
(raw ingestion)

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Silver Layer
(cleaning and transformations)

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Gold Layer
(business-ready tables)

4. Bronze Layer

Purpose:

- Store raw data
- Minimal transformations
- Preserve original information

Example:

Read JSON/CSV files and create a raw Delta table.

5. Silver Layer

Purpose:

- Clean data
- Remove duplicates
- Apply business rules
- Join datasets

Example:

Customer data cleansing and validation.

6. Gold Layer

Purpose:

- Analytics-ready data
- Aggregations
- Reporting tables

Example:

Daily sales summary.

7. Example Python DLT Code

```
from pyspark import pipelines as dp
```

```
@dp.table
def bronze_customer():
    return spark.read.format("json").load("/data/customer")
```

```
@dp.table
def silver_customer():
    return spark.sql(
        "SELECT * FROM bronze_customer WHERE customer_id IS NOT NULL"
    )
```

8. Streaming Tables

Streaming tables are used for incremental processing.

They are useful for continuously arriving data from sources like files or message systems.

9. Pipeline Modes

Triggered:

- Runs when started
- Good for batch schedules

Continuous:

- Keeps processing new data
- Good for near real-time workloads

10. DLT vs Normal Spark Jobs

Normal Spark Job:

- Developer controls workflow manually

DLT:

- Developer defines what tables should exist
- Databricks manages execution flow

11. Interview Points

Important topics:

- Bronze Silver Gold architecture
- Auto Loader with DLT
- Expectations and data quality
- Streaming tables
- Pipeline monitoring
- Unity Catalog integration

12. Summary

DLT helps data engineers create scalable, maintainable, and reliable data pipelines with less operational code.